

EDA USING POWER BI

WEEK - 1

REPORT

SUBMITTED BY

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Week 1:

Superstore Sales Data Import, Power Query & Data Cleaning

Dataset:

Superstore Sales Dataset

Problem Statement:

A retail organization wants to analyze its sales data to understand customer purchases, product categories, regional performance, and profitability. Before creating a business intelligence dashboard, the raw Superstore dataset must be imported, cleaned, transformed, and prepared in Power BI.

Objectives:

Import data into Power BI Desktop and navigate Report, Data, and Model views.

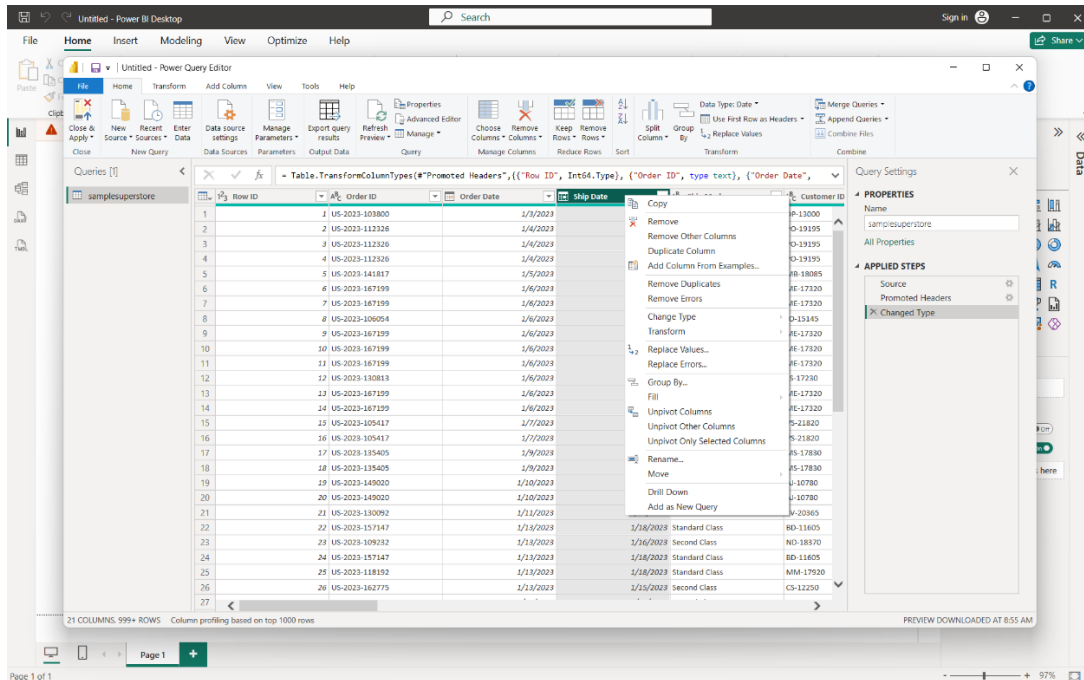
Inspect dataset attributes using Power Query Editor; verify data types, missing values, and record uniqueness.

Perform column transformations (rename, split, text trimming, data type coercion).

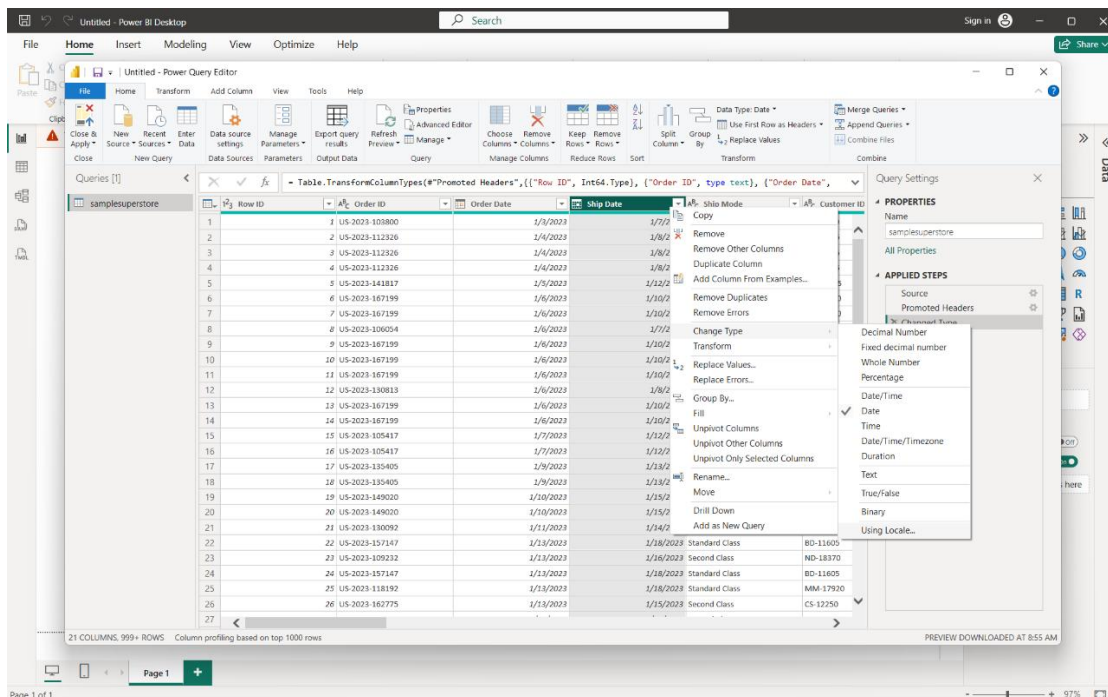
Extract temporal attributes from Order Date (Year, Quarter, Month, Day, Day of Week).

Tasks & Deliverables:

1. Load Superstore CSV into Power Query Editor.



2. Standardize text formatting across categorical attributes (Category, Sub-Category, Segment).



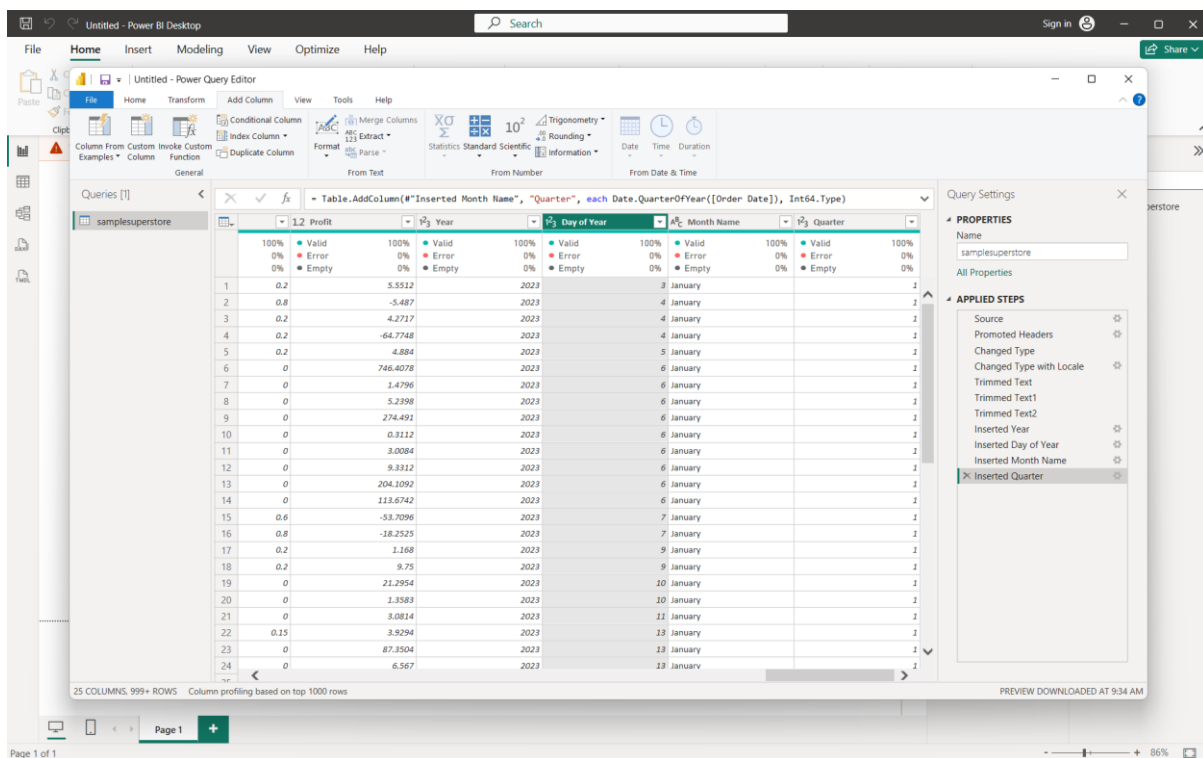
3. Perform column transformations (rename, split, text trimming, data type coercion).

The image displays two screenshots of the Microsoft Power BI Desktop interface, specifically the Power Query Editor, demonstrating data transformations.

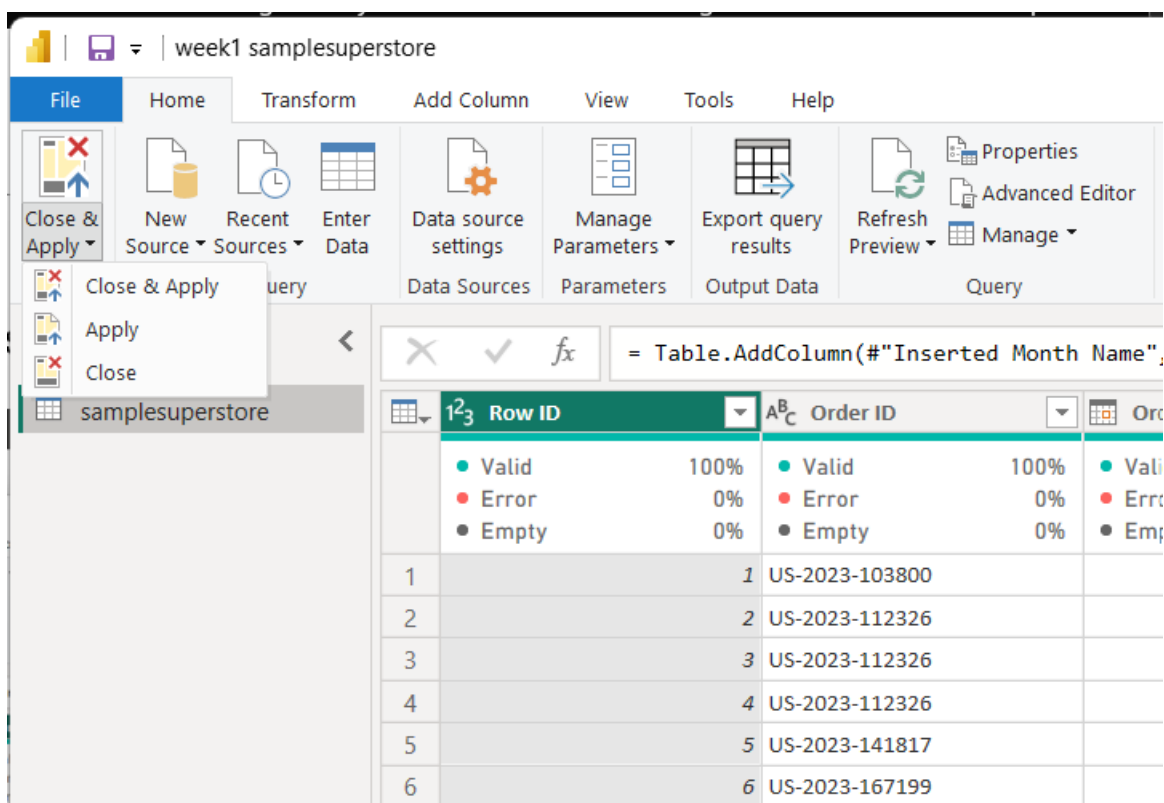
Top Screenshot: The query is named "Table.TransformColumnsType", with the formula bar showing: `Table.TransformColumnsType(*"Changed Type", {"Ship Date", type date}), "en-US"`. The data table has columns: Code, Region, Product ID, Category, Sub-Category, and Product Name. The "Region" column is highlighted in blue. The "Query Settings" pane on the right shows the "APPLIED STEPS" list, which includes "Changed Type with Locale".

Bottom Screenshot: The query is named "Table.TransformColumns", with the formula bar showing: `Table.TransformColumns(*"Changed Type with Locale", {"Order ID", Text.Trim, type text})`. The data table has columns: Order ID, Order Date, Ship Date, Ship Mode, and Customer ID. The "Order ID" column is highlighted in blue. The "Query Settings" pane on the right shows the "APPLIED STEPS" list, which includes "Trimmed Text".

4. Extract temporal attributes from Order Date (Year, Quarter, Month, Day, Day of Week).”



5. Close & Apply transformations to load a clean data model



Conclusion:

The Superstore Sales dataset was successfully imported and cleaned using Power Query in Power BI. The data types were verified, unnecessary inconsistencies were handled, and required column transformations were performed. Date-related attributes were also extracted from the Order Date column. The cleaned dataset is now ready for further analysis and visualization in Power BI.